THE BOUNDED CRITERION FOR THE CANONICAL MAP OF COMPOSED DERIVED FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7526-7543, following the statement at 7509-7524.

Editorial note. The construction of t is carried over from the proof of the general lemma by cross-reference; the isomorphism step is a direct application of lemma-leray-acyclicity.

Allowed standard prerequisites (editorial).

- Construction of t in the general case. lemma-compose-derived-functors-general (same chapter, cited at ~7535; its proof contains stated omissions and was not counted). Identifies the map t being tested.
- Acyclicity gives isomorphisms. lemma-leray-acyclicity (same chapter, cited at ~7542; PARTIAL item pending Chief ruling). The displayed arrow is an isomorphism.

Source reference. Original lemma 05T2: sources/stacks-derived/pinned-original-source.tex, source lines 5347–5347. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 015E: sources/stacks-derived/pinned-original-source.tex, source lines 5934–5934. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be left exact functors. Assume $\mathcal{A}$, $\mathcal{B}$ have enough injectives. The following are equivalent*

(1) *$F(I)$ is right acyclic for G for each injective object I of $\mathcal{A}$, and*
(2) *the canonical map*

$$t : R(G \circ F) \longrightarrow RG \circ RF.$$

is isomorphism of functors from $D^+(\mathcal{A})$ to $D^+(\mathcal{C})$.

Proof. If (2) holds, then (1) follows by evaluating the isomorphism t on $RF(I) = F(I)$. Conversely, assume (1) holds. Let $A^\bullet$ be a bounded below complex of $\mathcal{A}$. Choose an injective resolution $A^\bullet \to I^\bullet$. The map t is given (see proof of Lemma 05T2) by the maps

$$R(G \circ F)(A^\bullet) = (G \circ F)(I^\bullet) = G(F(I^\bullet))) \to RG(F(I^\bullet)) = RG(RF(A^\bullet))$$

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where the arrow is an isomorphism by Lemma 015E. □